

entail large-scale movement of freight traffic along the tracks of the Eastern and South Eastern Railways. This was beyond the power of steam traction and electrification seemed inevitable.

Initially, the scheme was drawn on the direct current system, based on the already-electrified Bombay and Madras sections. However, seeing the success of the alternating current system on the Société Nationale des Chemins de fer Français (SNCF) or the French National Railways, India decided to explore its feasibility for the main lines. Electrical engineering experts from the SNCF investigated the prospects of electrification on 25 KV AC 50 cycles Single Phase and recommended it. By 1958, the Indian Railways adopted the system because of operational and financial benefits. The first steps involved the electrification of the Eastern Railway's Bardhaman–Gomoh section, Igatpuri–Bhusaval on the Delhi–Bombay line of the Central Railway and Tambaram–Villupuram on the Southern Railway. It was extended from Gomoh in Jharkhand, to Mughalsarai in Uttar Pradesh, on the Eastern Railway. Priority was given to Asansol–Tatanagar and Rourkela and Rajkharswan–Barajamda on the South Eastern Railway. Until 1970, expansion was carried out rigorously in the coal–steel belt of the east. The focus then shifted to the Delhi–Bombay and Bombay–Ahmedabad lines of the Western Railway and the Jolarpettai–Olavakkode section of the Southern Railway.

Railway Minister Kengal Hanumanthaiya hailed the electric traction in his Railway Budget speech for 1971–72. He said: “Steam engines have lower speeds and acceleration, less thermal efficiency and hauling capacity ... they require coaling and watering facilities en route. Diesel and electric engines with higher speed, better acceleration, greater hauling capacity and less need for servicing en route are able to achieve improved operation and substantial savings in working expenses.”

Way forward

The complex process of electrification necessitated major overhauling of equipment and systems as well, such as signalling, telecommunication, and the import of new types of locomotives. This was, in a way, the beginning of a technological boom for the railways and paved the way for several initiatives. For instance, in a first in the world, the Diesel Locomotive Works (DLW) in 2018 successfully completed the conversion